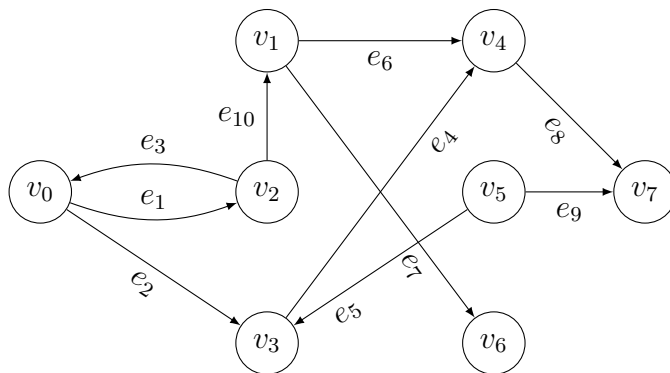


Graphs

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Practice quiz

Consider the following directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \Phi)$:



First, write the incidence function Φ of the directed graph, and check whether it is injective. Then, for each of the following graphs, say whether it is a subgraph of $\mathcal{G}$.

1. First graph: $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1, \Phi_1)$, where

$$\mathcal{V}_1 = \{v_0, v_2, v_3, v_5\},$$

$$\mathcal{E}_1 = \{a_1, a_2, a_3\},$$

$$\Phi_1(a_1) = (v_0, v_2),$$

$$\Phi_1(a_2) = (v_3, v_5),$$

$$\Phi_1(a_3) = (v_0, v_3).$$

2. Second graph: $\mathcal{G}_2 = (\mathcal{V}_2, \mathcal{E}_2, \Phi_2)$, where

$$\begin{aligned}\mathcal{V}_2 &= \{v_1, v_4, v_6, v_7\}, \\ \mathcal{E}_2 &= \{a_1, a_2, a_3\}, \\ \Phi_2(a_1) &= (v_1, v_4), \\ \Phi_2(a_2) &= (v_1, v_6), \\ \Phi_2(a_3) &= (v_4, v_7).\end{aligned}$$

3. Third graph: $\mathcal{G}_3 = (\mathcal{V}_3, \mathcal{E}_3, \Phi_3)$, where

$$\begin{aligned}\mathcal{V}_3 &= \{v_1, v_3, v_4, v_6\}, \\ \mathcal{E}_3 &= \{a_1, a_2, a_3, a_4\}, \\ \Phi_3(a_1) &= (v_1, v_4), \\ \Phi_3(a_2) &= (v_4, v_7), \\ \Phi_3(a_3) &= (v_3, v_4), \\ \Phi_3(a_4) &= (v_1, v_6).\end{aligned}$$